

Video Calls (Jitsi)



Video Calls (Jitsi)

In-app property tours with FCM ring, Jitsi rooms and server-signed tokens

Category: Real-time Communication | Backend: Jitsi (JaaS or self-hosted) | Status: Production

Jitsi Meet	Listing / Direct	FCM ring	WebView
Provider	Triggers	Push	Embed

1. What it is

WaveMart lets a buyer schedule an in-app video walkthrough of a property without ever leaving the platform. Calls are powered by Jitsi (JaaS for production, self-hosted community server for staging) and the in-app ring screen is triggered by an FCM data message, so a call works even when the user is on another screen.

2. How it works

01 Start from listing 1	GET /api/conferences/create/{listing} mints a Jitsi JWT and creates a conference row.
02 Invite buyer 2	POST /api/conferences/{id}/invite/{user} sends a push to the buyer with the conference id.
03 Buyer accepts 3	Tap on the FCM notification -> GET /api/conferences/{id}/join -> mints a join token.
04 Direct call 4	POST /api/conferences/start-direct/{conversation} opens a call from the chat screen.
05 WebView 5	flutter_inappwebview loads the Jitsi room URL with the user JWT.
06 Ping 6	POST /api/conferences/{id}/ping keeps the conference alive for late joiners.

3. Capabilities

Capability	What it does
One-tap walkthrough	Buyer requests a tour; seller gets a push; both meet in a Jitsi room.
Direct call from chat	Voice/video call from the conversation screen.
Server-minted JWTs	Backend signs Jitsi tokens so rooms are private and identity-bound.

FCM ring screen	Full-screen ring overlay shown above any other UI when an incoming call arrives.
Web session cookie	Optional web-session endpoint lets a desktop browser join a Jitsi room authenticated as the user.
Active call state	Ping endpoint, server-side status field (pending / active / ended).
Auto re-join on network	Jitsi client handles reconnection; the app can re-derive the join token.

4. Investor relevance

WHY IT MATTERS

In emerging markets, buyers want to verify the property before paying for travel. By embedding Jitsi directly, WaveMart becomes a property-touring app as well as a marketplace, dramatically shortening the time between interest and offer.

5. Tech surface

Component	Location
Conference service (mobile)	lib/data/services/conference_service.dart
WebView Jitsi screen	lib/presentation/screens/calls/webview_jitsi_screen.dart
Incoming call screen	lib/presentation/screens/calls/incoming_call_screen.dart
Jitsi service (backend)	WaveMart/app/Services/JitsiService.php
Conference controller	WaveMart/app/Http/Controllers/Api/ConferenceController.php